

Non-Additive Strain–Doping Coupling in Quasi-Hexagonal C₆₀ Graphullerene

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Substitutional doping and biaxial strain are jointly used to tune covalent molecular networks, yet total energies are still often assembled from separate composition-only and strain-only density-functional theory calculations. We study quasi-hexagonal C₆₀ graphullerene with the PBE+D3 functional and define a synergy order parameter $\mathcal{S}$ (meV/atom) for the total-energy cross term missing from sequential strain–doping scans. On B/N/P tetramers and periodic $n \times \text{C}_{60}$ supercells ($n = 1\text{--}8$), donor and acceptor dopants show opposite linear strain coefficients α , while phosphorus—despite the smallest tetramer $|\alpha|$ —exhibits the largest joint-load cross term, with $|\mathcal{S}|$ reaching 31.9 meV/atom at $n=1$ under +3% biaxial strain. These bounds are *rigid-strain upper bounds* at fixed fractional coordinates, not equilibrium thermodynamic quantities; ionic relaxation can reverse $\mathcal{S}$, and tetramer α is sensitive to dopant placement at $\sim 5\%$ loading. The results indicate that stability ranking under load in covalent molecular networks requires explicit coupled strain–doping energy evaluations when chemical mismatch or strain magnitude enter the geometric-nonlinear regime, rather than additive composition–strain models used in sequential high-throughput screening.

I. INTRODUCTION

Covalent molecular networks—from graphullerenes to covalent organic frameworks—are often tuned jointly by substitutional chemistry and mechanical load, yet many high-throughput pipelines still assemble $E_{\text{tot}}(\epsilon, \delta)$ from *separate* doping-only and strain-only density-functional theory runs [1–3]. Defect formation energies and substitutional stabilities under strain are classical questions in defect physics, but most workflows still assume that composition-only and strain-only increments superpose linearly—an assumption that breaks down when dopant-induced internal stress couples nonlinearly to external load. The omitted cross term is a systematic bias in sequential screening: it arises whenever a dopant introduces a *local chemical-mismatch stress field* that couples nonlinearly to an external biaxial load, so additive composition–strain models can mis-rank stability margins even when dopant ordering at $\epsilon=0$ is unchanged. Synthetic carbon allotropes with extended π conjugation underpin optoelectronic applications; charge transport in molecular networks is often polaronic and anisotropic [4–7], yet prior work emphasizes band edges, mobility, or hopping rather than a periodic total-energy cross term under joint biaxial (ϵ, δ) load. Quasi-hexagonal (qHP) C₆₀ graphullerene [4, 5] provides a covalently linked model network for such coupled tuning, with interfullerene linking that raises hopping couplings relative to vdW C₆₀ monolayers [4]; foundational and recent DFT surveys of qHP monolayers and covalent fullerene networks emphasize elastic, optical, and transport properties [8, 9] but do not report a periodic total-energy cross term under joint biaxial (ϵ, δ) load. Ab initio models of pristine qHP networks report intrinsic intramolecular strain from the eight-bond hexagonal connectivity [10] ($\sim 0.1\text{--}0.2$ eV-scale cage distortions relative to isolated C₆₀ in prior surveys), so external biaxial load couples to

an already pre-strained, *soft* molecular cage—a regime where Eshelby-type linear superposition of defect and external fields is not guaranteed. Beyond pristine monolayers, strain and substitution independently reshape band edges [2, 11, 12], while substitutional B/N engineering can yield spin-polarized half-semiconductor gaps [13], and related 2D fullerene networks show anisotropic mechanical and tribological response under load [14, 15] and endohedral qHP C₆₀ cage filling reshapes band gaps [16]; none of these reports a periodic total-energy cross term under joint biaxial (ϵ, δ) load, motivating explicit coupled grids rather than an additive superposition of composition and strain effects. Non-additive coupling is documented in other two-dimensional networks [1, 17]; a quantitative *total-energy* synergy order parameter $\mathcal{S}$ [Eq. (1)] has not been reported for substitutional B/N/P qHP C₆₀ tetramers and $n = 1\text{--}8$ supercells.

We address three questions on this *class* of covalent molecular networks (qHP C₆₀ as the validation platform): (1) how opposite linear strain coefficients α emerge for donor vs. acceptor dopants; (2) how large the omitted synergy cross term $|\mathcal{S}|$ is under joint load; and (3) whether $|\mathcal{S}|$ scales with size mismatch or instead reflects π -electron nonlinearity. Using $\gtrsim 90$ DFT tasks catalogued in Table II, we define $\mathcal{S}$ on periodic PBE+D3 grids (fifteen (n, δ) points at +3% strain). Matched-functional tetramer α and $\mathcal{S}_{+3\%}^{\text{tet}}$ on an independent placement map (Table V, PBE+D3) are compared to periodic $\mathcal{S}(n)$; the legacy-PBE reference column in Table VI is retained only as an archived mechanical baseline. The largest periodic $|\mathcal{S}|$ at fixed coordinates (up to 31.9 meV/atom for P at $n=1$) does not track tetramer $|\alpha|$ in any simple monotonic sense—*decoupling* here means that linear α cannot quantitatively predict cross-term magnitude or sign, not that $|\alpha|$ and $|\mathcal{S}|$ are anticorrelated. *Local* dopant distortion and *global* biaxial strain interact nonlinearly in total energy: $|\mathcal{S}|$ can exceed what

linear α alone would suggest, so *relative* stability margins from $\epsilon = 0$ substitution energies or strain-only elastic increments need not survive a coupled load even if they do not reorder dopants with widely separated E_{sub} values. Protocol limits (ionic-relaxation sign reversal, placement sensitivity) are benchmarked in Sec. IIF; polaron transport diagnostics appear in the Supplemental Material and do not enter $\mathcal{S}$.

II. COMPUTATIONAL METHODS

A. Electronic-structure setup

All periodic audit and charged-state tasks use CP2K [18] with PBE [19] and Grimme DFT-D3 (Becke–Johnson damping) [20]. Table VI lists an archived legacy-PBE tetramer grid (no D3); matched-functional PBE+D3 α on the *reference* placement map is being re-computed (open archive), while Table V already supplies a full PBE+D3 alternate-placement audit. Table II catalogues task groups, N_{conv} , main-text linkage, and protocol; superseded grids are in Table III. We do not employ the Koopmans-compliant rVV10 approach used in Ref. 4, and instead adopt a PBE+D3 protocol for periodic and charged-state tasks suitable for large-scale parameter scanning, at the cost of absolute accuracy in polaron binding energies.

B. Notation and conventions

Biaxial strain is denoted ϵ (percent); dopant species $\delta \in \{B, N, P\}$. The synergy order parameter is denoted by the calligraphic symbol $\mathcal{S}$ (never the Latin s), with units of meV per atom [Eq. (1)]. π -projected densities of states are written π -DOS (never bare “pi” in equations).

1. Rigid-strain protocol

Biaxial strain scales the in-plane lattice vectors of the *pristine* equilibrium cell at fixed fractional coordinates (no ionic relaxation under load). Reported α and $\mathcal{S}$ under this protocol are upper bounds at fixed coordinates (Sec. IIF).

2. Plane-wave cutoff (size scaling)

Archived CP2K inputs use 400 Ry for $n \leq 4$ and 350 Ry for $n \geq 6$ (Table II); comparisons at fixed n always share the same cutoff. Cross- n trends mix a plane-wave cutoff change at $n = 6$; the cutoff benchmark row in Table II provides a 400 Ry control at $6 \times \text{C}_{60}$ (Supplemental Material Table S1 for $n \geq 6$ production rows).

C. Substitutional doping and grids

Tetramer models ($4 \times \text{C}_{60}$, 240 sites) substitute B, N, or P at nominal 5% (12–13 sites from archived coordinates); these dopant configurations are not thermodynamically optimized ground states. A second tetramer placement ensemble (24 single-point energies: 18 dopant-strain points plus six pristine references; Table V) tests whether tetramer α and $\mathcal{S}$ are placement-sensitive at $\sim 5\%$ nominal doping. All periodic supercells propagate the same dopant site map (one substituent per C_{60}); quantitative $|\mathcal{S}|$ on the fifteen-point grid is therefore configuration-specific, whereas the alternate tetramer placement in Table V reverses the N α sign relative to the reference archive (+45 vs. -303 meV/%) and amplifies $|\alpha_P|$ by two orders of magnitude (+990 vs. -19 meV/%), so donor/acceptor α trends are placement-specific at $\sim 5\%$ nominal doping (Sec. IIE2). Periodic scaling cells place one substituent per C_{60} (n dopants among $60n$ lattice atoms, ~ 1.67 at.% for all n), contrasting with the tetramer grid ($\sim 5\%$ nominal). Counting a single dopant in the full supercell (e.g., $1/240 = 0.42$ at.% for the tetramer) is *not* the concentration used here; atomic percent is always dopants per lattice atom (n in $60n$). Tetramer α , gaps, and tetramer IPR/ J trends are not directly interchangeable with periodic $\mathcal{S}(n)$ without a matched-concentration benchmark; all $\mathcal{S}$ claims use the periodic protocol. In total, $\gtrsim 90$ tasks in Table II support the protocol-audited energy claims below (Marcus and Hirshfeld rows are orthogonal diagnostics). Charged-state tasks supply adiabatic IP/EA and vertical Marcus $\lambda^\pm$ (Supplemental Material; Sec. IIE6), which do not enter Eq. (1).

D. Energy observables

Linear strain sensitivity $\alpha = dE/d\epsilon$ (meV per 1% biaxial strain, tetramer *total* energy) is obtained from rigid tetramer strain grids (Table VI legacy PBE; Table V PBE+D3 alternate placement; reference PBE+D3 modernization pending). Periodic $\mathcal{S}$ and size-scaling energies use PBE+D3 throughout (Table II). Positive α means tension *raises* total energy; negative α means tension *lowers* it under the rigid-strain protocol. The raw substitution energy difference $\Delta E_{\text{sub}} = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ at fixed strain is *not* a thermodynamic standard formation enthalpy (no elemental chemical potentials); tabulated values are used only for *relative* ranking within this protocol (Table VI). In sequential high-throughput screens one often estimates $E_{\text{tot}}(\epsilon, \delta) \approx E_{\text{sub}}(\epsilon=0) + E_{\text{strain}}(\epsilon)$; the neglected term is precisely $\mathcal{S}(\epsilon, \delta)$ —the absolute error in that additive estimate. The order-of-magnitude ratio $|\mathcal{S}|/|E_{\text{sub}}|_{\text{per atom}}$ for screening workflows is discussed in Sec. IV as a rough magnitude estimate only (not a core nonlinearity test).

1. Non-additive coupling metric

We quantify strain–doping cross terms with the synergy order parameter $\mathcal{S}$:

$$\mathcal{S}(\epsilon, \delta) = \Delta E(\epsilon, \delta) - [\Delta E(\epsilon, 0) + \Delta E(0, \delta)], \quad (1)$$

where ϵ is biaxial strain, $\delta \in \{B, N, P\}$, and $\Delta E(\epsilon, \delta)$ is the energy per atom relative to the pristine, zero-strain reference at the same supercell size $n \times \text{C}_{60}$. The bracketed term is the *additive* prediction from separate strain-only and doping-only scans; deviations from zero measure non-additive coupling. Additive superposition gives $\mathcal{S} = 0$; $\mathcal{S} < 0$ indicates cooperative stabilization beyond the two-scan sum, whereas $\mathcal{S} > 0$ indicates cooperative destabilization.

When $\alpha_\delta = dE/d\epsilon$ denotes the *total*-cell linear strain coefficient (meV per 1% strain) at fixed n , a first-order estimate of the per-atom cross term is $\mathcal{S}(\epsilon, \delta) \approx (\alpha_\delta - \alpha_0)\epsilon/N_{\text{atoms}}$ at small ϵ under the rigid-strain protocol; tetramer and periodic n use different N_{atoms} and dopant concentrations, so this link is not used quantitatively in the periodic audit (Discussion, argument scope).

Periodic-cell $\mathcal{S}$ values are extracted from CP2K total energies (Table II); the fifteen-point +3% strain audit accompanies the open data release. We report $\mathcal{S}$ for total energy only. By the same four-corner construction one may define $\mathcal{S}_{\text{gap}} = E_g(\epsilon, \delta) - [E_g(\epsilon, 0) + E_g(0, \delta) - E_g(0, 0)]$ (and analogously $\mathcal{S}_J$, $\mathcal{S}_\lambda$ for hopping or reorganization energies); these observables are *not* tabulated here—panel (a) of Fig. 1 shows that gap slopes need not track $|\mathcal{S}|$.

a. Attribution framework (total energy). Conceptually, $\mathcal{S} = \mathcal{S}_{\text{geom}} + \mathcal{S}_{\text{el}} + \mathcal{S}_{\text{vdW}}$, where $\mathcal{S}_{\text{geom}}$ collects nonlinear elastic work from bond-angle and cage-shape changes beyond first-order strain, $\mathcal{S}_{\text{el}}$ from charge-transfer and orbital-rehybridization nonlinearities, and $\mathcal{S}_{\text{vdW}}$ from the Grimme D3 cross term. On the matched-functional tetramer P rigid four-corner grid, extracting D3 dispersion energies gives $\mathcal{S}_{\text{vdW}} \approx +0.33$ meV/atom—negligible versus $|\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}}| \approx 1.2$ meV/atom (Table IV). Full corner-resolved projections for B/N and periodic cells remain open; Table VII ($\bar{d}$, Δr_{cov}) and Hirshfeld population (Sec. IIE 4) proxy geometric vs. electronic channels in Discussion.

2. Localization and coupling (open data)

Following Capobianco *et al.* [4], inverse participation ratios and nearest-neighbor hopping J are archived for tetramer dimer fragments at the same PBE+D3 level as the energy scans; they are not used to infer $\mathcal{S}$.

3. Polaron binding

Marcus reorganization energies follow the vertical–adiabatic decomposition [4] (Supplemental Material).

Charged geometry optimizations supply adiabatic IP/EA; vertical single-points at the neutral geometry yield Marcus $\lambda^\pm$.

E. Detailed calculation protocols

1. DFT calculator settings

Periodic total-energy and electronic-structure tasks use CP2K [18] with PBE [19] and Grimme DFT-D3 (Becke–Johnson damping) [20]. Tetramer α and E_{sub} in Table VI come from the archived rigid tetramer grid (legacy PBE without D3). Unless stated otherwise, biaxial strain scales the in-plane lattice vectors of the pristine equilibrium cell at fixed fractional coordinates (rigid-strain protocol). The plane-wave cutoff is 400 Ry for $n \leq 4$ and 350 Ry for $n \geq 6$ (Table II); the atom-centered basis (DZVP-MOLOPT-SR-GTH) is unchanged across n . Tetramer models ($4 \times \text{C}_{60}$, 240 atoms) use B/N/P substitution at nominal 5% (12–13 dopant sites from archived coordinates). Periodic supercells place one substituent per C_{60} unit ($n = 1$ –8), giving n dopants among $60n$ lattice atoms (~ 1.67 at.% for all n).

2. Alternate-placement benchmark (Table V)

The *reference* placement in Table V is the archived rigid tetramer map underlying Table VI (legacy PBE without D3). The *alternate* placement repeats the full B/N/P strain grid ($\epsilon \in \{-5, -2.5, 0, +2.5, +3, +5\}\%$; 18 rigid single-point tasks) with an independently rerolled substitution map at the same nominal 5% doping level (pseudorandom seed 137; public archive). Reference-placement $\mathcal{S}_{+3\%}^{\text{tet}}$ for B/N/P is tabulated in Table V and Sec. IIIC; periodic $\mathcal{S}(n)$ is not inferred from either tetramer map. Alternate tasks use PBE+D3 rigid single-point total-energy evaluations at 400 Ry with an SCF convergence threshold of $10^{-6} E_h$; they are *not* on the same exchange-correlation treatment as the reference α column, so only within-placement trends are compared once the grid converges. At periodic $n=4$, two independent site maps repeat the four-corner Eq. (1) audit (pristine and B/N/P at $\epsilon=0$ and +3%; eight energies per map, 16 total) under the same PBE+D3 rigid protocol as the fifteen-point size-scaling grid; rankings of $|\mathcal{S}|$ at +3% are compared to the production reference map in Table I. Fixed pseudorandom seeds 137 and 271 identify the alternate maps in the archive; the production map uses seed 42. Where alternate-grid rigid single-points stall in SCF, the convergence threshold may be relaxed to $10^{-5} E_h$ for that task.

3. Ionic-relaxation benchmark (Table IV)

The four tetramer corners in Table IV are pristine and P -substituted cells at $\epsilon=0$ and $+3\%$ biaxial strain (the Eq. (1) grid for P). They use fixed-cell geometry optimization with BFGS (maximum 300 ionic steps for pristine corners and 450 for P -substituted corners) at prescribed in-plane cell vectors; only fractional coordinates relax while lattice vectors remain fixed. BFGS stops at 300 steps or when the maximum and RMS force components fall below 4.5×10^{-4} and 3.0×10^{-4} Ha/bohr, or when the maximum and RMS atomic displacements fall below 3.0×10^{-3} and 1.5×10^{-3} bohr, respectively. These tasks use the same PBE+D3 settings as periodic audits: 400 Ry plane-wave cutoff, DZVP-MOLOPT-SR-GTH basis, and an SCF convergence threshold of $10^{-6} E_h$ (pristine) or $10^{-5} E_h$ (P -substituted) with OT-DIIS (up to 300 inner SCF iterations per ionic step and 20–40 outer SCF cycles) before updating the density for the next BFGS displacement. Rigid reference energies in Table IV remain the archived legacy-PBE (no D3) single-point grid; relaxed corners use PBE+D3, so the legacy cross-functional row is sign-qualitative only. A matched-functional control recomputes the same four corners as rigid PBE+D3 single points at the relaxed fractional coordinates (400 Ry, $10^{-6} E_h$ SCF, no ionic steps); quantitative retention is read from the matched-functional row in Table IV, while the legacy-PBE rigid row remains for comparability with the archived tetramer grid.

4. Population strain path ($n=1$ P)

Six rigid single-point tasks on the $n=1$ P -substituted periodic supercell scan Hirshfeld atomic charges along $\epsilon \in \{-5, -2.5, 0, +2.5, +3, +5\}\%$ (Table II); Mayer bond orders are extracted in post-processing from converged wavefunctions. These tasks use the same PBE+D3, 400 Ry, and $10^{-6} E_h$ SCF settings as the periodic size-scaling grid; the neutral $n=1$ P cell has an odd electron count, so spin-polarized KS (LSD .TRUE.) is enabled (unlike the closed-shell size-scaling ENERGY tasks). They complement geometric $\bar{d}$ metrics in Table VII.

5. Periodic ionic-relaxation benchmark (pending)

Fixed-cell geometry optimization on the periodic $n=1$ P -substitution four corners (pristine and P at $\epsilon=0$ and $+3\%$; PBE+D3, 400 Ry) uses the same BFGS thresholds as Table IV. Inputs are generated from the converged rigid single-point cells in the size-scaling archive; results will be tabulated separately once all four corners converge (do not rescale tetramer Table IV benchmarks).

6. Marcus vertical step and post-processing

Charged-state geometry optimizations supply adiabatic ionization potentials and electron affinities ($E_{\text{opt}}^{\pm}$). Eight vertical single-point energy evaluations compute $E_{\text{vert}}^{\pm}$ at the *neutral*-geometry coordinates with the charged UKS settings of the relaxed state, following Ref. 4; $\lambda^{\pm} = E_{\text{vert}}^{\pm} - E_{\text{opt}}^{\pm}$ (Supplemental Material). For near-gap B doping, negative λ^{-} from the vertical step at fixed neutral coordinates is treated as a semilocal artifact and excluded from rate estimates. π -projected densities of states use a 0.05 eV Gaussian broadening aligned to the Fermi level (Fig. 2). Inverse participation ratios and hopping integrals J are defined in Sec. II; verified IPR and J values accompany the open data release (not used to infer S).

F. Validation benchmarks

The task-group inventory in Table II (superseded exploratory grids in Table III) lists N_{conv} and main-text linkage; ionic-relaxation energies appear in Table IV; the alternate-placement grid in Table V spans the full B/N/P strain path and six modern PBE+D3 pristine references. Three validation benchmarks—ionic-relaxation tetramer S_P (Table IV), alternate dopant placement (Table V), and a cutoff benchmark at $6 \times C_{60}$ N @ $+3\%$ (Table II)—govern interpretation of size-scaling trends (Secs. III C and IV).

Table VI and Sec. III B report E_{sub} using the substitution-energy definition above (no chemical-potential reference; not a formation enthalpy).

III. RESULTS

We organize the results around total-energy non-additivity (Sec. II B); Secs. III A–III C report band-edge context, tetramer α and geometry (Tables VI and VII), and periodic $S(n)$ (Tables II and I). Polaron transport diagnostics (Marcus λ , IPR, J) are in the Supplemental Material and are orthogonal to the S audit.

A. Electronic structure under strain and doping

Figure 1(a,b) summarizes tetramer HOMO–LUMO gaps (archived tetramer electronic-structure set) and representative π -DOS (B @ $\epsilon=0$; color scheme follows Ref. 4). At $\epsilon=0$, B is near a closed π gap ($E_g \approx 0.03$ eV), N doping pins the Fermi level inside the conduction band ($E_g \approx -0.14$ eV), and P remains gapped ($E_g \approx 0.05$ eV); tensile strain narrows the pristine gap (~ 23 meV/% on the three available strain points in the dataset). Donor vs. acceptor ordering of the $\epsilon=0$ gaps tracks the opposite α signs in Sec. III B, but band-edge slopes are not

TABLE I. Periodic synergy audit at +3% biaxial strain (PBE+D3 rigid protocol, reference placement seed 42; core $n \leq 4$ at 400 Ry). $\mathcal{S}$ in meV/atom from Eq. (1). $n \geq 6$ rows (350 Ry production settings) are in Supplemental Material Table S1.

n (C_{60} units)	Dopant	ϵ (%)	$\mathcal{S}$ (meV/atom)
1	B	+3	-8.38
1	N	+3	+2.30
1	P	+3	-31.91
2	B	+3	-2.18
2	N	+3	+6.85
2	P	+3	-28.29
4	B	+3	-6.23
4	N	+3	+2.95
4	P	+3	-23.71

TABLE II. DFT task groups and main-text linkage (~ 90 tasks supporting quantitative claims). Superseded exploratory grids are listed in Table III.

Task group	System	N_{conv}	Main text	Protocol
Strain response (α , E_{sub})	$4 \times C_{60}$	24	Table VI	Rigid strain; legacy PBE
π -DOS and gaps	$4 \times C_{60}$	12	Fig. 2	PBE+D3
IPR (polaron dimer)	Dimer	4	Data release	Pristine/B; $\epsilon=0$, +3%
Hopping J	Dimer	4	open data	PBE+D3; same corners
Charged polarons ^a	$4 \times C_{60}$	20	Suppl. Mat.	UKS; 12 GEO + 8 vert. SP
Periodic size scaling	$n \times C_{60}$	41	$\mathcal{S}(n)$	PBE+D3; 400/350 Ry
Cutoff benchmark	$6 \times C_{60}$ N	1	Sec. IIF	+3%; 400 Ry control
Ionic relaxation	$4 \times C_{60}$	4	Table IV	Fixed-cell GEO; P corners
Alternate placement	$4 \times C_{60}$	24	Table V	Rigid PBE+D3; alternate map
Periodic placement	$4 \times C_{60}$	16	Table I	$n=4$; two independent maps
Matched-functional rigid	$4 \times C_{60}$	4	Table IV	PBE+D3 @ relaxed coords
Hirshfeld population	$1 \times C_{60}$ P	6	Discussion	Spin-polarized KS

^aSix relaxed tetramer geometries provide charged-state initial structures (not counted in N_{conv}).

used to infer $\mathcal{S}$. The four-panel π -DOS stack (Fig. 2) shows N-induced states crossing the Fermi level at $\epsilon=0$ (metallic pinning), a narrow π gap for B ($E_g \approx 0.03$ eV), and a wider gap for P—consistent with excess vs. deficient π electron count at the dopant cage but not yet decomposed into site-resolved Mulliken or Bader partitions (Limitations).

B. Mechanical strain sensitivity

Pristine tetramers are strain-insensitive ($|\alpha| \approx 1$ meV/%). On the archived legacy-PBE reference placement (Table VI), substitution opens an order-of-magnitude spread in α : N (-303.2 meV/%, tension stabilizes the doped cell), B ($+56.9$ meV/%, tension destabilizes), and P (-19.1 meV/%, weakly stabilizing). Matched-functional PBE+D3 α on the alternate placement map (Table V) gives $+21.4$, $+45.4$, and $+989.6$ meV/% for B, N, and P; Fig. 1(c) uses this PBE+D3 column so mechanical slopes are comparable to periodic $\mathcal{S}$. The opposite sign for N vs. B persists across the $\epsilon \in \{-5, \dots, +5\}$ % grid at this fixed placement and is not an artifact of a single strain point; Table V confirms placement-sensitive slopes (N α sign reversal on the alternate map). At $\epsilon = +5\%$, N remains

the most tension-stabilized dopant (α most negative), whereas B shows the strongest destabilization under tension; P sits between these extremes with a modest negative α , foreshadowing why its linear coefficient poorly predicts $|\mathcal{S}| = 23.7$ meV/atom for P at $n=4$ under +3% strain (Sec. III C). Substitution energies at $\epsilon = 0$ span $E_{\text{sub}} = +73.4$ eV/dopant (B), -115.3 eV/dopant (N), and -15.8 eV/dopant (P)—raw total-energy differences without chemical-potential reference, not standard formation enthalpies—with $n_{\text{dop}} = 12$ –13 counted from input coordinates (Table VI). Figure 1(c) highlights the α split; ranking dopants solely by $|E_{\text{sub}}|$ at $\epsilon=0$ is misleading once +3% strain is applied. Table VII links mechanical spread to *tetramer* local geometry ($\sim 5\%$ nominal doping, multiple sites): P shows negligible strain-induced change in the mean nearest-neighbor C–C distance ($\Delta \bar{d} \approx 3 \times 10^{-5}$ Å vs. $\sim 8.5 \times 10^{-3}$ Å for B at +3% load), indicating a vanishing *uniform nearest-neighbor stretch* component rather than absence of cage buckling; periodic $n=1$ bond statistics are not yet tabulated (Limitations). Complementary Hirshfeld population on the periodic $n=1$ P strain path (Table II; Sec. IIE 4) probes charge redistribution along the same load path as Table VII geometry.

TABLE III. Superseded exploratory DFT grids (excluded from the ~ 90 -task count in Table II).

Task group	System	N_{conv}	Disposition
Early strain probe	$2 \times \text{C}_{60}$	5	Superseded
Single-molecule doping	C_{60}	11	Superseded
Early electronics	$2 \times \text{C}_{60}$	24	Superseded
Exploratory design grid	$4 \times \text{C}_{60}$	20	Archive only

TABLE IV. Ionic-relaxation benchmark for the tetramer P -substitution synergy grid (four fixed-cell geometry-optimization corners: pristine and P at $\epsilon=0$ and $+3\%$, same biaxial cell). This benchmark is *sign-qualitative only*: the rigid reference uses legacy PBE without D3, whereas relaxed corners use PBE+D3, so the legacy cross-functional row is *not* a quantitative retention ratio. A matched-functional rigid reference (PBE+D3, same four corners): $\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}} = +1.19$ meV/atom; $|\mathcal{S}_{\text{relaxed}}|/|\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}}| \approx 1.9$ (sign *not* preserved on the matched-functional grid). We test whether $\mathcal{S}$ retains *sign within the tetramer grid only*; it does *not* calibrate periodic main-text $\mathcal{S}(n)$, which uses different dopant concentration and boundary conditions. Rigid reference energies: archived fixed-coordinate tetramer single-point grid (legacy PBE, no D3). Relaxed energies: PBE+D3 fixed-cell geometry optimization (400 Ry) from archived inputs in the open repository. $\mathcal{S}$ follows Eq. (1) with $N=240$ atoms.

Quantity	Rigid (legacy tetramer)	Relaxed (geometry opt.)
$E(\text{pristine}, \epsilon=0) (E_h)$	-1363.5512	-1363.9551
$E(\text{pristine}, \epsilon=3\%) (E_h)$	-1363.5522	-1363.9550
$E(P, \epsilon=0) (E_h)$	-1370.4965	-1374.4415
$E(P, \epsilon=3\%) (E_h)$	-1370.4890	-1374.4615
$\mathcal{S}$ (meV/atom)	+0.96	-2.28
Sign of $\mathcal{S}$ vs. rigid	+	- (not preserved)

C. Non-additive strain-doping coupling

Figure 1(d) tracks protocol $\mathcal{S}(n)$ at $+3\%$ biaxial strain. At $n = 4$ under the fixed periodic production map (Table I; two independent $n=4$ placement maps preserve the reference-map $|\mathcal{S}|$ ranking at $+3\%$: $P > B > N$), $\mathcal{S} = -6.2$ meV/atom (B), $+3.0$ meV/atom (N), and -23.7 meV/atom (P); P shows the largest magnitude despite the smallest $|\alpha|$ on the legacy tetramer grid (Table VI). On the alternate tetramer map (Table V, PBE+D3 throughout), P still carries the largest $\mathcal{S}_{+3\%}^{\text{tet}}$ ($+41.4$ meV/atom) while $|\alpha|$ is placement-dominated ($+989.6$ meV/%); this is *not* a simple $\alpha-|\mathcal{S}|$ anticorrelation—decoupling means linear α cannot predict cross-term magnitude or sign (Sec. IV D). For $n \leq 4$ at matched 400 Ry cutoffs, $|\mathcal{S}|$ reaches 31.9 meV/atom for P at $n = 1$ and remains $\mathcal{O}(10)$ meV/atom for B and P at $n = 2-4$ (Table I). Within $n \leq 4$, $|\mathcal{S}|$ is not monotonic in n (e.g., B: $8.4 \rightarrow 2.2 \rightarrow 6.2$ meV/atom at $n=1, 2, 4$), reflecting finite-cell dopant-dipole and image interactions rather than an asymptotic size limit; we report protocol values at each supercell without extrapolating $\mathcal{S}_\infty$. Rows with $n \geq 6$ use 350 Ry production settings and appear only in Supplemental Material Table S1. All periodic $\mathcal{S}$ values quoted in this subsection use the fixed dopant-site map and are tabulated in Table II (fifteen $n \times$ dopant points at $+3\%$ strain). On the reference tetramer placement, legacy PBE rigid $\mathcal{S}_{+3\%}^{\text{tet}}$ from Eq. (1) is $+3.8$, $+5.7$, and $+1.0$ meV/atom for B, N, and P (Table V); the alternate placement column (Table V) uses PBE+D3 rigid single-points and is not used to infer periodic $\mathcal{S}(n)$. Negative $\mathcal{S}$ (B, P at $n = 4$) means coupled tension *stabilizes*

the doped cell relative to the additive estimate; positive $\mathcal{S}$ (N at $n = 4$) means sequential screens *overestimate* stability under load.

1. Ionic-relaxation checkpoint (Table IV)

Table IV benchmarks four fixed-cell geometry-optimization corners on the tetramer P -substitution grid (pristine and P at $\epsilon=0$ and $+3\%$). For the pristine tetramer, relaxed total energies at $\epsilon=0$ and $+3\%$ are -1363.9551 and -1363.9550 E_h ($\Delta E \approx 1.2 \times 10^{-4}$ E_h), roughly an order of magnitude smaller than the archived rigid legacy-PBE nominal strain increment ($\sim 10^{-3}$ E_h). The P -doped corners give $E(P, \epsilon=0) = -1374.4415$ and $E(P, \epsilon=3\%) = -1374.4615$ E_h . Because rigid references use legacy PBE without D3 whereas relaxed corners use PBE+D3 (Sec. II E 3), the cross-functional comparison gives $\mathcal{S}_{\text{rigid}} = +0.96$ meV/atom versus $\mathcal{S}_{\text{relaxed}} = -2.28$ meV/atom (sign *not* preserved); matched-functional PBE+D3 rigid single-points at the same relaxed coordinates yield $\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}} = +1.19$ meV/atom with the same sign reversal ($|\mathcal{S}_{\text{relaxed}}|/|\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}}| \approx 1.9$; Table IV). This tetramer benchmark does not rescale periodic $\mathcal{S}(n)$ (concentration and boundary conditions differ; Sec. IV).

TABLE V. Alternate dopant-placement sensitivity (*tetramer only*, rigid strain). The *reference* placement defines main-text tetramer α (Table VI) and the periodic supercell site map (one substituent per C_{60}); its α and $\mathcal{S}_{+3\%}^{\text{tet}}$ use the archived legacy-PBE grid (no D3). The *alternate* placement repeats the full B/N/P strain grid ($\epsilon \in \{-5, -2.5, 0, +2.5, +3, +5\}\%$; 18 rigid single-points) with PBE+D3 on an independently rerolled substitution map (Sec. II E 2); alternate rows report verified placement sensitivity at $\sim 5\%$ nominal doping without mixing periodic boundary conditions. $\mathcal{S}_{+3\%}^{\text{tet}}$ follows Eq. (1) on the archived rigid tetramer four-corner grid ($N=240$). On the alternate map, large $|\alpha_P|$ and large $|\mathcal{S}_{+3\%}^{\text{tet}}|$ coexist—decoupling is *not* an α - $|\mathcal{S}|$ anticorrelation (Sec. IV D). Periodic $\mathcal{S}(n)$ is *not* tabulated here because concentration and boundary conditions differ; the independent $n=4$ periodic placement audit (Table II) and the alternate *tetramer* grid (full B/N/P strain path plus six modern PBE+D3 pristine references) are reported in Sec. II F.^a

Dopant	Placement	α (meV/%)	$\mathcal{S}_{+3\%}^{\text{tet}}$ (meV/atom)	Status
B	reference	+56.9	+3.8	verified (Table VI)
N	reference	-303.2	+5.7	verified
P	reference	-19.1	+1.0	verified
B	alternate	+21.4	-2.6	verified (alternate)
N	alternate	+45.4	+12.6	verified (alternate)
P	alternate	+989.6	+41.4	verified (alternate)

^a Alternate site map: pseudorandom seed 137 (public archive).

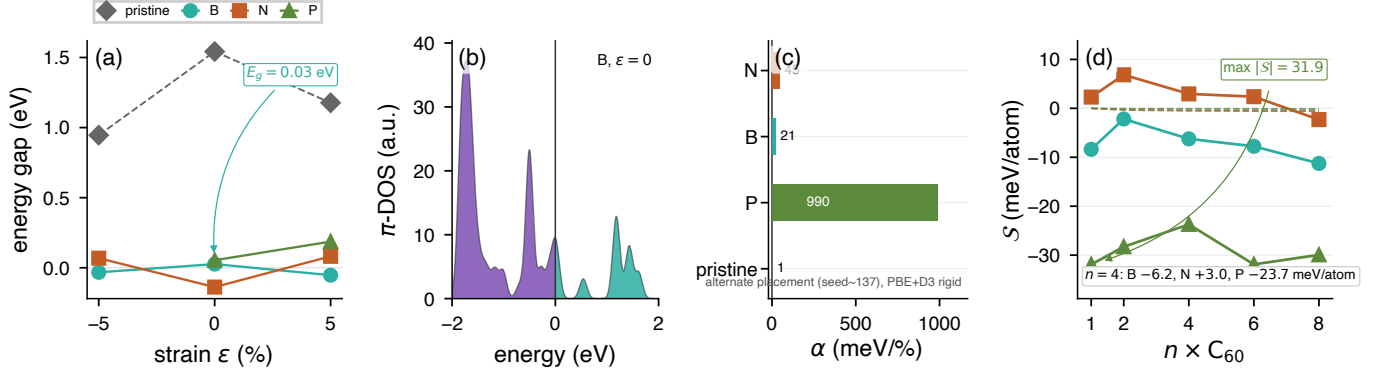


FIG. 1. Coupled (ϵ, δ) response in doped qHP C_{60} . (a) HOMO-LUMO gap vs. strain (tetramer electronic archive; PBE). (b) π -DOS for B @ $\epsilon=0$ (PBE+D3; N panel: $E_g < 0$ indicates metallic pinning). (c) Mechanical α (PBE+D3 rigid tetramer; alternate placement, Table V; legacy reference column in Table VI). (d) Periodic $\mathcal{S}(n)$ at +3% biaxial strain (PBE+D3 rigid; $n \leq 4$ core grid; peak $|\mathcal{S}|$ at P, $n=1$).

IV. DISCUSSION

A. Literature context

Relative to strain-only, doping-only, mobility, or mechanical studies in qHP and related fullerene networks [1, 2, 4, 7–16, 21, 22], the distinction here is a periodic *total-energy* audit of joint biaxial (ϵ, δ) load [Eq. (1)] (periodic PBE+D3; Table II). Foundational DFT surveys of qHP C_{60} monolayers emphasize anisotropic elastic moduli and optical gaps [8] rather than matched four-corner total-energy grids under joint substitutional and mechanical load. Strain-engineering roadmaps for generic two-dimensional materials [3] likewise foreground band-edge and mobility levers under separate mechanical or chemical protocols rather than a matched four-corner total-energy grid on covalent fullerene networks. Graphendofullerene derivatives report Curie temperatures tunable through *either* biaxial strain or electrostatic doping applied in isolation [12], reinforcing that

joint (ϵ, δ) total-energy corners are not implicit in sequential protocol designs. In other 2D hosts, substitutional defects can raise or lower formation energies under biaxial strain depending on dopant–host size mismatch [22]; spin-resolved impurity engineering on qHP graphullerene instead targets half-semiconductor gaps [13]; we instead quantify the omitted additive cross term $\mathcal{S}$ in total energy. Band-edge metrics contextualize donor/acceptor ordering (Sec. III A) but are not used to infer $\mathcal{S}$.

B. Internal stress and external strain

Substitutional B, N, or P can be viewed as embedding a *localized internal strain field* ϵ_{int} into an already distorted qHP cage baseline [10]. Table VII quantifies the chemical-mismatch scale by covalent-radius excess Δr_{cov} (+7, -6, and +30 pm for B, N, and P vs. C) and by mean dopant–C distance $\bar{d}$ at $\epsilon=0$. External biaxial strain ϵ_{ext} is a uniform macroscopic load on the periodic cell. Se-

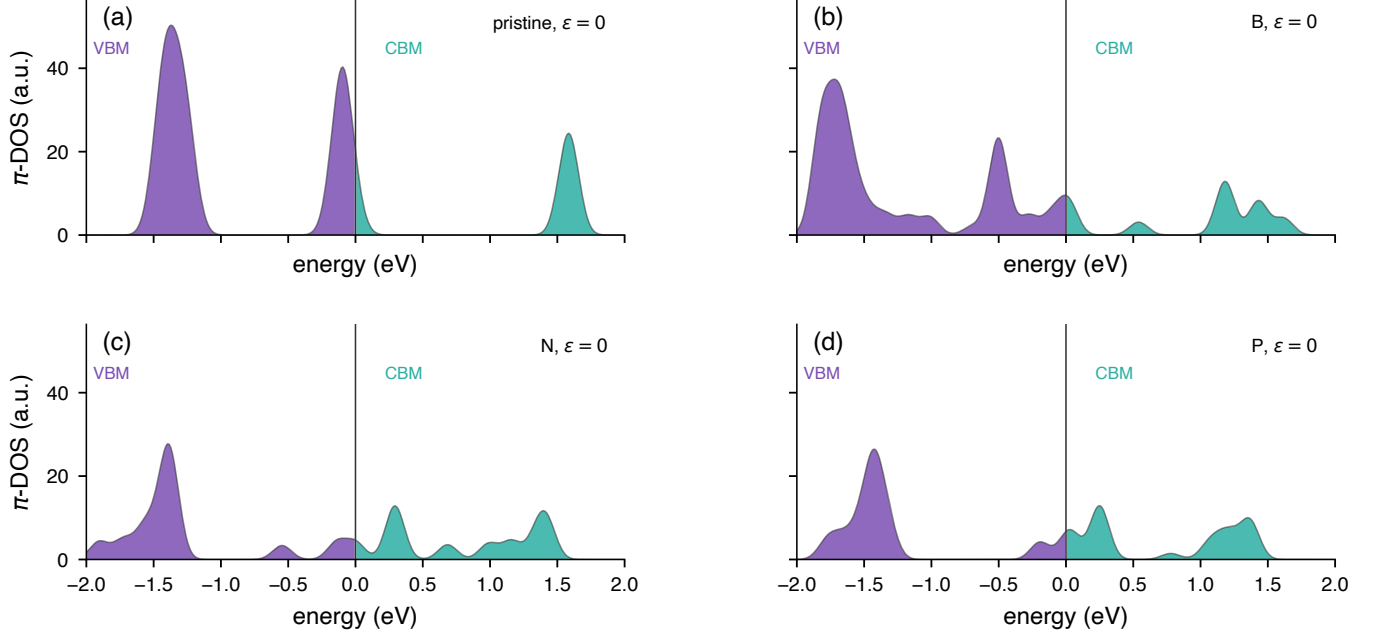


FIG. 2. π -projected density of states (π -DOS) for pristine and B/N/P-doped tetramers at $\epsilon=0$ (PBE+D3). Panels (a)–(d) share a common vertical scale; energy is relative to the Fermi level at 0 eV (Capobianco color scheme [4]).

TABLE VI. Linear strain sensitivities and raw substitution energy differences for tetramer models (archived rigid-strain grid, legacy PBE without D3). Tabulated ΔE_{sub} values are *not* thermodynamic formation enthalpies (no chemical potentials); use only for relative ranking within this protocol. Matched-functional PBE+D3 α on the alternate placement map appears in Table V; reference-placement PBE+D3 modernization is in progress. $E_{\text{sub}} = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ at $\epsilon=0$ with no chemical-potential reference (not a standard formation enthalpy). n_{dopants} is counted from archived input coordinates (nominal 5% target; B: 13, N: 12, P: 12). **Not a formation enthalpy:** E_{sub} is a fixed-protocol total-energy difference per dopant site with no elemental chemical-potential reference; use only for *relative* ranking within this study, not standard thermodynamic formation energies.

System	n_{dop}	$E(\epsilon=0) (E_h)$	$E(\epsilon=+5\%) (E_h)$	α (meV/%)	E_{sub} (eV/dopant)
Pristine	—	−1363.5512	−1363.5509	−1.0	—
B-doped	13	−1328.4900	−1328.4090	+56.9	+73.4
N-doped	12	−1414.3812	−1414.4053	−303.2	−115.3
P-doped	12	−1370.4965	−1370.4888	−19.1	−15.8

quential high-throughput models implicitly assume that responses to ϵ_{int} and ϵ_{ext} superpose linearly; $\mathcal{S}$ is the total-energy evidence that this superposition fails when both fields are present. In the Eshelby picture, a small substitutional defect in a linear elastic continuum produces a strain field that superposes with external load; qHP C₆₀ violates that limit because the cage is *soft* and pre-strained [10], so second-order defect–elastic work (our geometric channel) can exceed linear $\alpha \times \epsilon$ scaling.

For *size-mismatched* P, the dopant introduces compressive local misfit in the cage ($\Delta r_{\text{cov}} = +30$ pm) while tensile ϵ_{ext} partially relieves that misfit without proportionally stretching nearest-neighbor bonds ($\Delta \bar{d} \approx 3 \times 10^{-5}$ Å at +3%; Table VII). The resulting negative $\mathcal{S}$ (cooperative stabilization relative to the additive estimate) is consistent with nonlinear *defect–elastic* coupling: external strain couples to *pre-existing cage buck-*

ling rather than to uniform bond stretching. For B and N, $|\Delta r_{\text{cov}}| \lesssim 10$ pm and $\Delta \bar{d}$ tracks ϵ_{ext} more uniformly ($\sim 8.5 \times 10^{-3}$ Å at +3%), so $\mathcal{S}$ is set primarily by *electronic* nonlinearity—donor vs. acceptor π counting and Fermi-level pinning (Secs. III A and III B)—rather than by large-amplitude cage reshaping. In this language, small-mismatch dopants approximate the linear-additive regime ($|\mathcal{S}| \sim \mathcal{O}(1\text{--}10)$ meV/atom at $n=1$), whereas large-mismatch substituents require explicit coupled (ϵ, δ) corners because geometric nonlinearity drives $|\mathcal{S}|$ to $\mathcal{O}(10^2)$ times the per-atom strain increment from α alone (Sec. III C).

a. Component attribution. Full decomposition into $\mathcal{S}_{\text{geom}}$, $\mathcal{S}_{\text{el}}$, and $\mathcal{S}_{\text{vdW}}$ awaits corner-resolved post-processing on every four-corner grid. For the matched-functional tetramer *P* rigid grid, extracting Grimme D3 dispersion energies at the four corners gives

TABLE VII. Local dopant–cage structure from archived *tetramer* XYZ ($\sim 5\%$ nominal doping; qualitative proxy for mismatch trends, not periodic $n=1$ bond statistics). Mean nearest-neighbor C distance $\bar{d}$ in Å; standard deviation $\sigma(\bar{d})$ across dopant sites; covalent-radius excess vs. C in pm. Strain-induced change $\Delta\bar{d} = \bar{d}(\epsilon=+3\%) - \bar{d}(\epsilon=0)$ tracks the *uniform nearest-neighbor stretch* component; cage buckling and angle distortions are not captured by $\bar{d}$ alone (Discussion Secs. IV B and IV C 0 g).

Dopant	$\bar{d}(\epsilon=0)$ (Å)	$\bar{d}(\epsilon=+3\%)$ (Å)	$\Delta\bar{d}$ (Å)	$\sigma(\bar{d})$ @ +3% (Å)	Δr_{cov} (pm)
B	1.384	1.393	+0.0085	0.019	+7
N	1.393	1.402	+0.0087	0.024	−6
P	1.384	1.385	+0.00003	0.0004	+30

$\mathcal{S}_{\text{vdW}} \approx +0.33$ meV/atom—orders of magnitude below $|\mathcal{S}_{\text{rigid}}^{\text{PBE+D3}}| \approx 1.2$ meV/atom (Table IV)—so vdW non-additivity does not drive the reported cross terms. Within the present audit, P combines negligible $\Delta\bar{d}$ with the largest $|\mathcal{S}|$ and a monotonic Hirshfeld shift on the $n=1$ strain path ($+0.063e \rightarrow +0.056e$ at +3%; Sec. IV C 0 g), implicating *pre-stored cage strain plus electronic redistribution* rather than uniform bond stretching as the dominant channel. B and N Hirshfeld strain-path data are queued (Sec. II E 4); larger $\Delta\bar{d}$ with smaller $|\mathcal{S}|$ is consistent with electronic-dominated cross terms at small Δr_{cov} .

C. Mechanistic synthesis

a. Opposite α for N vs. B. N introduces excess π electrons that strengthen bonding under tension (negative α), whereas B creates electron deficiency and antibonding π^* character that rises with lattice expansion (positive α). Quantitatively, the $\epsilon=0$ gap ordering ($E_g^N < E_g^B < E_g^P$ within PBE; Sec. III A) and the dopant-resolved π -DOS stack in Figs. 1(b) and 2 (N states at E_F ; B near-gap; P gapped) align with donor vs. acceptor π counting, while total-energy α spans ~ 360 meV/% between N and B on the tetramer grid (Table VI). The opposite N vs. B sign is reported for this fixed placement across $\epsilon \in \{-5, \dots, +5\}\%$ (Table VI); Table V repeats the full tetramer strain grid on an alternate site map (PBE+D3 rigid; Sec. II E 2): N α changes sign and P $|\alpha|$ grows by $\mathcal{O}(10^2)$ relative to the reference placement, whereas $\mathcal{S}_{+3\%}^{\text{tet}}$ at +3% remains $\mathcal{O}(10)$ meV/atom for all three dopants on the alternate map. Mayer bond order and Bader charge partitioning along the strain path are not yet computed (Limitations); periodic $\mathcal{S}(n)$ magnitudes on the reference map are not placement-averaged, though two independent $n=4$ placement maps preserve the P>B>N ranking (Table II).

b. Large $|\mathcal{S}|$ for P despite small $|\alpha|$. P carries the largest covalent-radius excess (+30 pm vs. C; Table VII) and pre-existing cage distortion at $\epsilon=0$, yet shows the smallest strain-induced change in mean nearest C distance ($\Delta\bar{d} \approx 3 \times 10^{-5}$ Å vs. $\sim 8.5 \times 10^{-3}$ Å for B/N) and the smallest site-to-site spread at +3% ($\sigma(\bar{d}) \approx 4 \times 10^{-4}$ Å vs. $\sim 2 \times 10^{-2}$ Å for B/N; Table VII). Tensile load therefore couples to *pre-existing* cage distortion from size mismatch rather than to a uniform elastic response, yield-

ing a cross term invisible in separate scans but dominant when both strain and substitution are applied simultaneously—consistent with P’s intermediate E_{sub} yet the largest $|\mathcal{S}|$ at $n=4$ (23.7 meV/atom; Table II) despite weak tetramer $|\alpha|$. Sec. III B presents this Table VII geometric decoupling before the $n=4$ $\mathcal{S}$ ranking in Sec. III C: tetramer $|\alpha|$ ranks P as weakly strain-sensitive, whereas the coupled periodic audit exposes the largest cross term.

c. N S sign change with n. With one substituent per C_{60} , electrostatic fields dilute at large n . The tetramer linear coefficient α and the periodic cross term $\mathcal{S}$ probe different energy partitions: α is the slope of *total-cell* energy under uniform lattice expansion at $\sim 5\%$ multi-site doping, whereas $\mathcal{S}$ is the omitted four-corner cross term per atom at ~ 1.67 at.% under periodic boundary conditions. They therefore need not share sign even for the same element—a negative α (tension lowers total energy in a strain-only scan) can coexist with positive $\mathcal{S}$ (the coupled corner is *less* stable than an additive model predicts), as for N at $n=4$. Any N sign flip between $n=6$ and $n=8$ is confounded with the 400→350 Ry cutoff step and is excluded from design rules (Sec. II F). Together, these three classes suggest a taxonomy for covalent molecular networks: *electronic* donor/acceptor control of α , *structural* local-distortion coupling to external strain (P), and *finite-size* electrostatic dilution (N at large n).

d. Geometric vs. electronic nonlinearity. For P, Table VII shows that rigid tensile load barely moves the mean dopant–C distance ($\Delta\bar{d} \approx 3 \times 10^{-5}$ Å) yet $\mathcal{S}$ remains the largest at $n=4$ (23.7 meV/atom), whereas B/N exhibit $\sim 10^2 \times$ larger $\Delta\bar{d}$ at +3% but smaller $|\mathcal{S}|$ (6.2 and 3.0 meV/atom). At $n=4$ the coupled correction for P exceeds the rigid per-atom strain increment from tetramer α by two orders of magnitude ($|\mathcal{S}| \gg \alpha \times 3\%/N_{\text{atoms}}$; Methods energy observables), even though $|\alpha|$ ranks P as the weakest strain responder on the tetramer grid.

e. Argument scope (concentration and boundary conditions). We do *not* validate $\mathcal{S}$ by extrapolating tetramer α to periodic $n=4$: the $\sim 5\%$ tetramer vs. ~ 1.67 at.% periodic concentrations and tetramer vs. periodic boundary conditions are unmatched, so a first-order $(\alpha_\delta - \alpha_0)\epsilon/N_{\text{atoms}}$ estimate is excluded from the argument. Non-additivity is assessed only through Eq. (1) on matched four-corner grids at common (ϵ, δ) ; P in particular requires explicit coupled evaluation rather than

sequential α screening.

f. Sign origins of $\mathcal{S}$. For size-mismatched P (geometric channel), tensile ϵ_{ext} partially relieves dopant-induced cage compression; coupled corners are *more* stable than additive models predict, giving $\mathcal{S} < 0$ when the cross term is negative in our sign convention (B and P at $n=4$). For small-mismatch B and N (electronic channel), $\mathcal{S}$ follows nonlinear evolution of π states near E_F under strain: donor N and acceptor B occupy different filling regimes, so $\mathcal{S}$ can be positive or negative at comparable $|\Delta r_{\text{cov}}|$ (N +3.0 vs. B -6.2 meV/atom at $n=4$) without a single geometric scaling law.

g. Hirshfeld charge along the $n=1$ P strain path. On the periodic $n=1$ P grid (PBE+D3), Hirshfeld charge on P decreases from +0.063 e at $\epsilon=0$ to +0.056 e at +3% (+0.050 e at +5%) while tetramer $\Delta \bar{d}$ for P remains below 10^{-4} Å at +3% load (Table VII). Tensile strain therefore redistributes electron density without proportional nearest-neighbor bond stretching, consistent with coupling to pre-existing cage distortion rather than uniform elastic response; Mayer bond orders along the same path are not yet tabulated (Limitations).

h. Significance versus DFT uncertainty. $\mathcal{S}$ is a difference of total energies computed with the *same* functional, basis, and cell for all four corners of Eq. (1), so systematic errors largely cancel in sign and relative magnitude. As a *rough magnitude estimate only* (not a core result), at $n=4$ and +3% strain P has $|\mathcal{S}| = 23.7$ meV/atom while the tetramer P per-atom substitution scale is $|E_{\text{sub}}| n_{\text{dop}} / n_{\text{atoms}} \approx 0.79$ eV/atom ($|E_{\text{sub}}| = 15.8$ eV/dopant, $n_{\text{dop}} = 12$ on the 240-atom tetramer, Table VI)—i.e. $|\mathcal{S}| / |E_{\text{sub}}|_{\text{per atom}} \sim 3\%$ —but tetramer ($\sim 5\%$) and periodic $n=4$ (~ 1.67 at.%) concentrations differ and Table VI is not on the same functional as periodic $\mathcal{S}$. Large absolute uncertainties in PBE E_{sub} therefore do not invalidate the conclusion that sequential scans omit a meV/atom-scale cross term.

i. Decomposition outlook. The stress-field picture (Sec. IV B) and Table VII–Hirshfeld trends indicate that $\mathcal{S}$ is not reducible to uniform bond stretching; Mayer/Bader corner projections along B/N strain paths remain open (Limitations).

D. Generality and design principles

a. Decoupling definition. α - $\mathcal{S}$ decoupling does *not* assert anticorrelation between $|\alpha|$ and $|\mathcal{S}|$: on the alternate PBE+D3 map (Table V), P has both the largest $|\alpha|$ and the largest $\mathcal{S}_{+3\%}^{\text{tet}}$. The claim is that linear strain slopes from a one-parameter scan cannot quantitatively predict four-corner cross-term magnitude or sign because α is placement-sensitive at $\sim 5\%$ loading whereas $|\mathcal{S}|$ remains $\mathcal{O}(10)$ meV/atom on matched-functional tetramer and periodic audits.

b. Mismatch- $|\mathcal{S}|$ scaling (audit at $n=1$, +3%). Using $|\Delta r_{\text{cov}}|$ from Table VII as a single mismatch coordinate, periodic $|\mathcal{S}|$ spans 2.3–8.4 meV/atom for B and

N ($|\Delta r_{\text{cov}}| \lesssim 10$ pm) but reaches 31.9 meV/atom for P ($|\Delta r_{\text{cov}}| = 30$ pm; Table I). N breaks a naive “ $|\mathcal{S}| \propto (\Delta r)^2$ ” law at comparable mismatch to B because its cross term is electronic (positive $\mathcal{S}$ at $n=4$) rather than geometric. The data nonetheless support a *two-regime* design rule for covalent networks: $|\Delta r_{\text{cov}}| \gtrsim 20$ pm flags the geometric-nonlinear regime where coupled audits are mandatory (P); $|\Delta r_{\text{cov}}| \lesssim 15$ pm leaves $|\mathcal{S}|$ at few-meV/atom under +3% load but still permits sign conflicts between α and $\mathcal{S}$ (B, N). When $|\mathcal{S}|$ approaches tens of meV/atom—as for P at $n=1$ –4—stability margins inferred from $\epsilon=0$ substitution energies or strain-only α can shift by several percent of the per-atom substitution scale even without reordering dopants (Sec. IV C 0 h).

c. When additive scans suffice. Dopant mismatch: at $|\Delta r_{\text{cov}}| \lesssim 10$ pm and +3% biaxial load, $|\mathcal{S}| \lesssim 10$ meV/atom on the $n=1$ audit grid; sequential pipelines may be adequate for coarse ranking but not for close E_{sub} competitors. **Strain magnitude:** pristine tetramers show $|\alpha| \approx 1$ meV/%, so sub-1% loads keep linear strain increments small; at +3% and above, coupled corners for size-mismatched dopants are advisable because $|\mathcal{S}|$ can exceed the rigid per-atom increment from $\alpha \times \epsilon$ by orders of magnitude (P at $n=4$). **Network architecture:** qHP C₆₀ combines interfullerene covalent links with *soft* intramolecular cages and intrinsic intramolecular strain [10], amplifying local-distortion coupling relative to rigid 2D lattices; extrapolation to graphene-like hosts requires separate benchmarks (Limitations).

d. Co-design implications. At $n=4$ and +3% strain the coupled correction is $|\mathcal{S}| = 23.7$ meV/atom for P vs. 6.2 meV/atom for B—sequential *additive* estimates differ by approximately a factor of four in the load-dependent *margin* even though widely separated E_{sub} ranks at $\epsilon=0$ (e.g., B vs. P; Table VI) are unchanged. Dopants with covalent-radius excess $\gtrsim 20$ pm relative to C (P: +30 pm vs. B: +7 pm; Table VII) show the largest protocol $|\mathcal{S}|$ under fixed coordinates; *size-mismatched* substituents should be screened with coupled (ϵ, δ) total-energy evaluations rather than sequential one-parameter pipelines. The fifteen-point periodic grid tabulates protocol $|\mathcal{S}|$ at each (n, δ) as an audit of additive omission (Table II), not as quantitative design targets. At +3% strain, P carries the largest additive-approximation error in protocol $\mathcal{S}$ ($|\mathcal{S}| = 23.7$ –31.9 meV/atom from $n=4$ to $n=1$ under fixed coordinates); qualitative co-design should treat P under tension as a *coupled* stability problem (Table V shows $|\alpha_P|$ placement-dominated on the alternate map), not $|E_{\text{sub}}|$ at $\epsilon = 0$ alone. Where $\mathcal{S} < 0$ (B and P at $n=4$), coupled tension can stabilize doped cells beyond additive predictions; where $\mathcal{S} > 0$ (N at $n=4$), sequential screens overestimate stability under load. For B at $n=4$, negative $\mathcal{S}$ coexists with positive α —a sign conflict that strain-only elastic estimates cannot resolve without evaluating $\mathcal{S}$. For N at $n=4$, negative tetramer α and positive $\mathcal{S}$ present a second sign conflict at ~ 1.67 at.% periodic doping. Gap closing [Fig. 1(a)] does not track $|\mathcal{S}|$; $\mathcal{S}_{\text{gap}}$

is defined in Sec. II but not computed, so photonic or transport targets cannot be inferred from total-energy $\mathcal{S}$ alone.

E. Experimental signatures

Three classes of laboratory observables could test the stress-field picture without resolving total energies directly. **Raman or IR cage modes:** under joint doping and strain, mode shifts should deviate from the sum of doping-only and strain-only shifts when $|\mathcal{S}|$ is large—especially for P, where tensile load couples to pre-existing cage distortion rather than uniform bond stretching (Table VII). **Scanning tunneling spectroscopy:** dopant-site local density of states should evolve nonlinearly with strain when Hirshfeld charge redistributes without proportional $\Delta\bar{d}$ (Sec. IV C 0g); N pinning near E_F vs. gapped B/P provides contrasting signatures. **Thermal or chemical stability:** for $\mathcal{S} < 0$ (B, P at $n=4$), tension should enhance apparent doping stability relative to additive thermochemical estimates; for $\mathcal{S} > 0$ (N at $n=4$), the opposite bias is expected. Quantitative links require matched sample chemistry and strain calibration; we quote DFT protocol bounds only.

F. Validation protocol (Tables II–V)

Sec. III C 1 reports the tetramer ionic-relaxation benchmark (Table IV). The alternate placement map (Table V, PBE+D3 rigid grid) gives $\alpha = +21.4$, $+45.4$, and $+989.6$ meV/% and $\mathcal{S}_{+3\%}^{\text{tet}} = -2.6$, $+12.6$, and $+41.4$ meV/atom for B, N, and P; N α reverses sign relative to the legacy reference column (Table VI). Two independent $n=4$ periodic placement maps preserve the production-map $|\mathcal{S}|$ ranking at $+3\%$ (P>B>N; Table II). The cutoff benchmark row in Table II tests 400 Ry vs. 350 Ry for $6 \times \text{C}_{60}$ N @ $+3\%$.

G. Limitations and outlook

Rigid-strain periodic $\mathcal{S}(n)$ and legacy-PBE tetramer α (Table VI) are not on the same exchange-correlation grid; the PBE+D3 alternate tetramer map (Table V) and periodic PBE+D3 $\mathcal{S}$ audit supply matched-functional comparisons for placement and decoupling claims. Fixed-cell ionic relaxation is reported for the tetramer P grid (Table IV); periodic $n=1$ P four-corner relaxations are queued (PBE+D3; open archive) to translate rigid upper bounds into equilibrium periodic cross terms. All periodic $\mathcal{S}(n)$ use a single production dopant-site map; magnitudes are configuration-specific (two independent $n=4$ placement maps preserve the P>B>N ranking; Table II). Production $n \geq 6$ points use 350 Ry while core claims require matched 400 Ry (cutoff benchmark in Table II). Additional limits: tetramer vs. periodic

concentration mismatch; Table V tetramer geometry is a qualitative proxy; reference-placement PBE+D3 α modernization pending ('reference_pbed3/'); PBE + D3withoutKoopmans/*rVV10*; $\mathcal{S}_{\text{gap}}$ not quantified; Mayer bond orders not tabulated; Hirshfeld population covers $n=1$ P only (B/N queued; Sec. II E 4); unphysical B λ^- under the fixed neutral-geometry vertical protocol (Supplemental Material) is excluded from rate estimates; mobility tensors are not computed. Future work adds periodic ionic-relaxation corners (in progress), reference-placement matched-functional Table I, B/N Hirshfeld paths, Mayer/Bader strain-path partitioning, and placement averaging on periodic cells.

V. CONCLUSIONS

We quantify how substitutional chemistry and biaxial strain couple *nonlinearly* in covalent molecular networks: the omitted total-energy cross term $\mathcal{S}$ arises when a dopant-induced internal stress field meets an external load, not from a graphullerene-specific numerical artifact. On qHP C_{60} as the validation platform, two mechanisms emerge—*electronic* donor/acceptor control of α (B, N) and *geometric* local-distortion coupling for size-mismatched P—with a practical mismatch threshold ($|\Delta r_{\text{cov}}| \gtrsim 20$ pm) separating few-meV from tens-of-meV $|\mathcal{S}|$ under $+3\%$ load (Secs. IV B and IV D). In detail:

1. opposite donor/acceptor α for N vs. B on the archived tetramer grid (Table VI);
2. meV/atom-scale omitted cross terms on a fifteen-point periodic grid ($|\mathcal{S}|$ up to 31.9 meV/atom for P at $n=1$ at fixed coordinates; Table II);
3. $|\mathcal{S}|$ tracking local cage distortion and mismatch (Table VII) rather than tetramer $|\alpha|$ for P, including on the PBE+D3 alternate tetramer map (Table V).

Under the rigid-strain protocol (Sec. II B), quoted $|\mathcal{S}|$ magnitudes bound equilibrium cross terms until periodic ionic-relaxation corners complete (Table IV; queued $n=1$ P). Coupled (ϵ, δ) screening is advisable for size-mismatched substituents, $+3\%$ and larger loads, and close substitution-energy competitors; Raman, STS, and stability trends in Sec. IV E offer experimental tests of the stress-field picture.

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DATA AVAILABILITY

The data that support the findings of this study are openly available at <https://github.com/chenxingqiang/graphullerene-strain-engineering>, including computational inputs and outputs, tabulated synergy and polaron analyses, and materials to reproduce all figures and tables in the main text.

Ionic-relaxation, alternate tetramer placement, periodic placement, Hirshfeld population, and size-scaling archives referenced in Tables II–V are included in the public release. Fixed pseudorandom seeds identifying alternate site maps (e.g. seed 137 for Table V) are recorded in the archive metadata. A tagged release matching the manuscript revision will be issued upon acceptance.

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- [1] S. Chen, H. Liu, and K. Wang, untangling the respective effects of heteroatom doping and strain on electronic transport in 2D materials, *Advanced Materials* **36**, 2301234 (2024).
 - [2] H. Li, M. A. Shakoori, and R. Wang, strain engineering of electronic structure and thermoelectric properties of quasi-hexagonal fullerene monolayer, *J. Appl. Phys.* **136**, 015101 (2024).
 - [3] A. K. Katiyar and J. H. Ahn, strain-engineered 2D materials: challenges, opportunities, and future perspectives, *Small Methods* **9**, 2401404 (2025).
 - [4] A. Capobianco, A. Breglia, P. Gentile, G. Capano, A. Asinari, G. Drera, L. Sangaletti, M. Cossi, and S. Casassa, electron localization and mobility in monolayer fullerene networks, *Nano Lett.* **24**, 7891 (2024).
 - [5] Y. Yang, W. Zhang, H. Liu, and K. Li, two-dimensional carbon allotropes with tunable direct band gaps and high carrier mobility, *Appl. Surf. Sci.* **536**, 147815 (2021).
 - [6] P. Silva, R. M. Faria, L. G. Ferreira, and J. L. F. Fatima, large polarons in two-dimensional fullerene networks: the crucial role of anisotropy in charge transport, *Nanoscale* **16**, 2337 (2024).
 - [7] Y.-B. Shi, S.-H. Lv, Z.-F. Shao, H.-K. Dong, S. Cao, and P. Qian, a first-principles study of 1d and 2d C₆₀ nanostructures: strain effects on band alignments and carrier mobility, *J. Phys.: Condens. Matter* **35**, 225701 (2023).
 - [8] R. M. Tromer, L. A. R. Junior, and D. S. Galvao, A DFT study of the electronic, optical, and mechanical properties of a recently synthesized monolayer fullerene network, *Chemical Physics Letters* **804**, 139925 (2022).
 - [9] R. Lv, J. Xi, M. Chen, and S. Yang, Covalent fullerene networks: preparation, properties and catalytic applications, *Rev. Mater. Res.* **2**, 100195 (2026).
 - [10] G. Makov and U. Argaman, structure and properties of graphullerene: a semiconducting two-dimensional C₆₀ crystal, *npj Comput. Mater.* **9**, 211 (2023).
 - [11] B. Peng and M. Pizzochero, monolayer C₆₀ networks: a first-principles perspective, *Chem. Commun.* **61**, 10287 (2025).
 - [12] D. López-Alcalá, Z. Hu, and J. J. Baldoví, graphendo-fullerene: a novel molecular two-dimensional ferromagnet, *Chemical Science* **16**, 7659 (2025).
 - [13] M. A. Khan and M. N. Leuenberger, tuning the electronic and optical properties of impurity-engineered two-dimensional graphullerene half-semiconductors, *Phys. Rev. Mater.* **9**, 034001 (2025).
 - [14] J. Wang, K. Li, and H. Zhang, simulation on stress-related anisotropy of qTP C₆₀ and qHP C₆₀: implications for optoelectronic nanodevices, *ACS Applied Nano Materials* **7**, 3456 (2024).
 - [15] F. Qiu, H. Wang, P. Zhang, and Y. Li, atomic-scale insights into the mechanical and tribological properties of graphullerene, *Tribology International* **204**, 110756 (2025).
 - [16] M. L. P. Junior, R. M. Tromer, L. A. R. Junior, and D. S. Galvão, endohedral derivatives of the recently synthesized two-dimensional fullerene networks: electronic and optical insights from first-principles calculations, *arXiv preprint* (2026), arXiv:2603.10142 [cond-mat.mtrl-sci].
 - [17] M. Alihosseini and M. Neek-Amal, Strain tuning of optoelectronic properties of covalent organic framework bilayers and heterostructures, *Phys. Rev. B* **108**, 165403 (2023).
 - [18] J. VandeVondele, M. Krack, F. Mohamed, M. Parrinello, T. Chassaing, and J. Hutter, Quickstep: Fast and accurate density functional calculations using a mixed gaussian and plane waves approach, *Computer Physics Communications* **167**, 103 (2005).
 - [19] J. P. Perdew, K. Burke, and M. Ernzerhof, Generalized gradient approximation made simple, *Phys. Rev. Lett.* **77**, 3865 (1996).
 - [20] S. Grimme, S. Ehrlich, and L. Goerigk, effect of the damping function in dispersion corrected density functional theory, *Journal of Computational Chemistry* **32**, 1456 (2011).
 - [21] A. Shaikh, J. Wu, and B. Peng, negative and positive anisotropic thermal expansion in 2D fullerene networks, *arXiv preprint* (2025), arXiv:2504.02037 [cond-mat.mtrl-sci].
 - [22] P. Santra, S. Ghaderzadeh, M. Ghorbani-Asl, H.-P. Komsa, E. Besley, and A. V. Krashenninnikov, strain-modulated defect engineering of two-dimensional materials, *npj 2D Materials and Applications* **8**, 33 (2024).